Simulating the Chernobyl disaster

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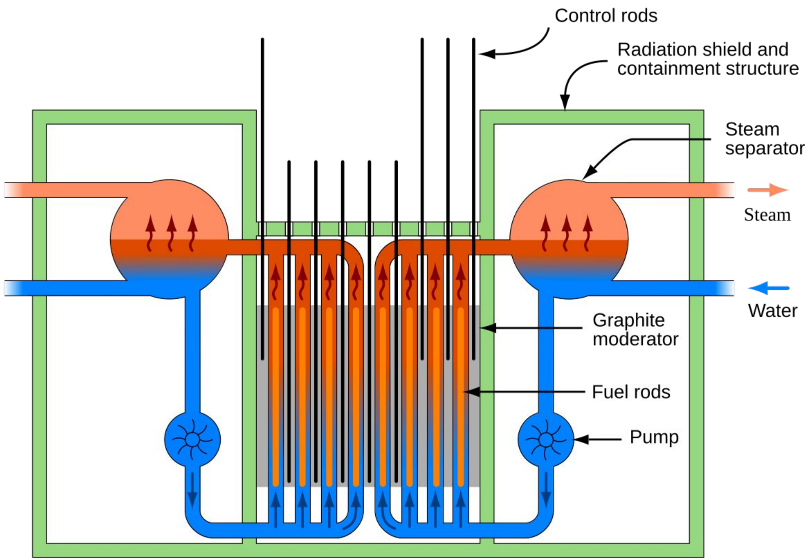
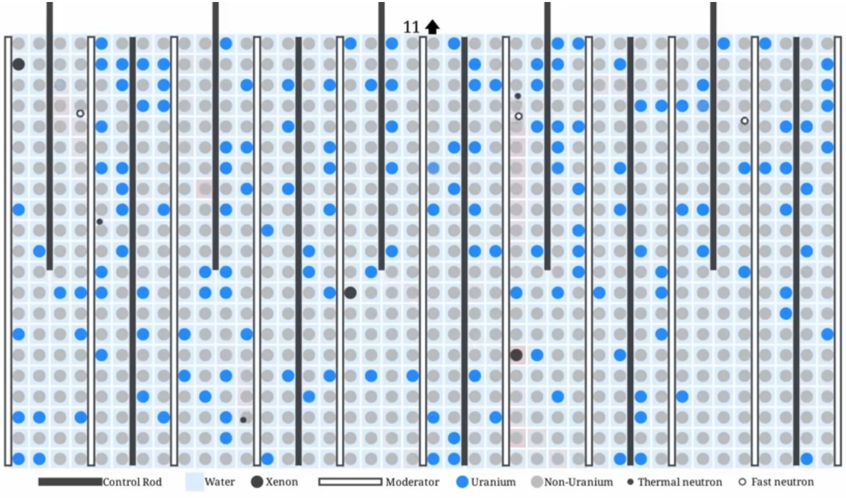
The RBMK reactor

- Graphite-moderated, water-cooled reactor;
- U-235 fission releases neutrons and heat;
- Boron control rods absorb neutrons and reduce power;
- Xe-135 absorbs neutrons strongly and can cause low-power instability;
- Model focuses on neutron balance and reactor feedback.



Source: https://en.wikipedia.org/wiki/RBMK#/media/File:RBMK_reactor_schematic.svg

Our inspiration vs final model

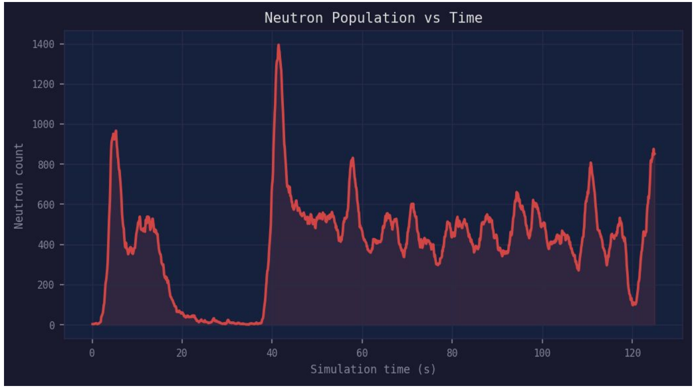


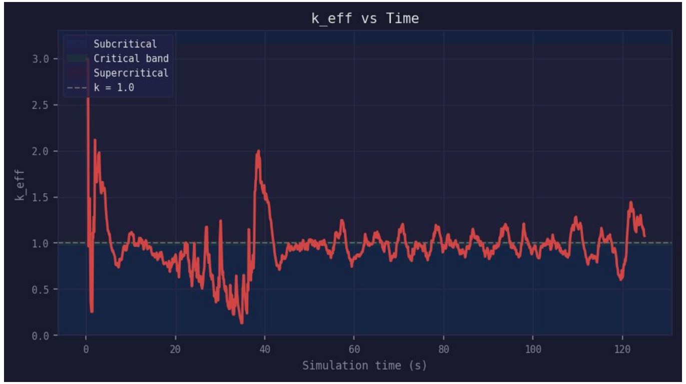
Demonstration

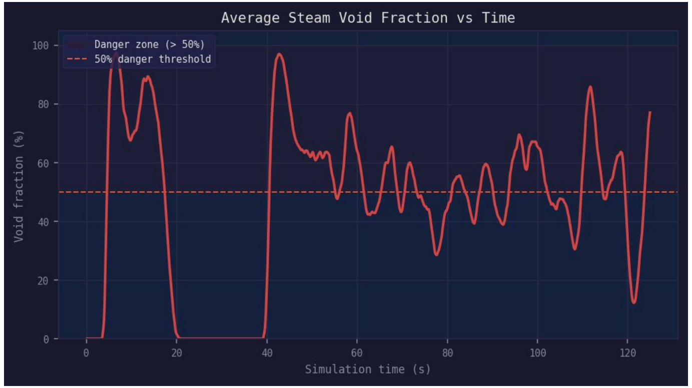
Demonstration of the model

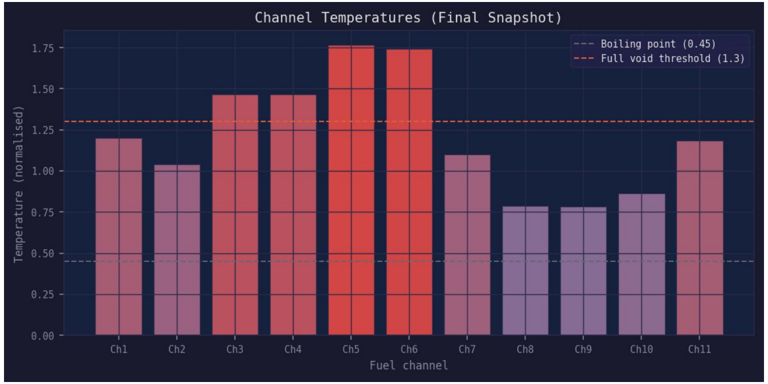
Demonstration of the Chernobyl Simulation

Simulation plots









Sources and reference values

- https://www-pub.iaea.org/MTCD/publications/PDF/Pub913e_web.pdf - World Nuclear Association, RBMK Reactors appendix.
- https://world-nuclear.org/information-library/appendices/rbmk-reactors - EPJ N (2021), "A simplified analysis of the Chernobyl accident": control-rod insertion reactivity `+396 pcm` with xenon poisoning vs `-344 pcm` without.
- https://www.epj-n.org/articles/epjn/full_html/2021/01/epjn200018/epjn200018.html
- YouTube animation as visual/storytelling inspiration: https://www.youtube.com/watch?v=P3oKNE72EzU
- Timeline of events: https://www.chernobylgallery.com/chernobyl-disaster/timeline/
- Similar works (collection of articles): https://www.frontiersin.org/research-topics/36576/advanced-modeling-and-simulation-of-nuclear-reactors/magazine
- Advanced topics (book): https://www.sciencedirect.com/book/monograph/9780128150696/modelling-of-nuclear-reactor-multi-physics

Thank you for your attention

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